Digital Data Transmission 2026–27

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Test 2. Decoding

Notation: the repetition code sends one information bit n times, and the ML and MAP decoders are those of the lecture notes. Where a bit is sent, the coded bit 0 is the symbol $-A$ and the coded bit 1 is the symbol $+A$ on the Gaussian channel.

1. A code sends k information bits in n channel uses. Its rate $R = k/n$ is measured in

(a) information bits per channel use. ⋆
(b) channel uses per information bit.

> *Solution:* The answer is (a). By Definition 2.1, the rate is the number of information bits carried by each channel use; the repetition code of length n has $R = 1/n$.

2. For a received sequence, the ML decoder chooses the codeword that

(a) was the most probable before the transmission.
(b) makes the received sequence most likely. ⋆

> *Solution:* The answer is (b). By Theorem 2.2, the ML decoder maximizes the likelihood of the received sequence. Only the MAP decoder uses the priors, and the two coincide for equiprobable messages.

3. On the bit-flipping channel with flip probability p, where $0 < p < \frac{1}{2}$, the repetition code of length 3 is used and the receiver observes 011. The ML decoder decides the bit

(a) 1. ⋆
(b) 0.

> *Solution:* The answer is (a). By Example 2.4(b) the ML decoder is the majority vote, since one flip is more likely than two: 011 is one flip away from 111 and two from 000.

4. On the real Gaussian channel the repetition code of length 3 is used and the receiver observes $\boldsymbol{y} = \left(2A, -\frac{A}{2}, -\frac{A}{2}\right)$. The ML decoder decides the bit

(a) 0, because two of the three outputs are negative.
(b) 1, because the sum of the outputs is positive. ⋆

> *Solution:* The answer is (b). By Example 2.4(a), the ML decoder adds the outputs and decides by the sign of the sum, here $u = A > 0$. A majority of the signs is the hard receiver of Section 2.3, which decides 0 on this output.

5. On the real Gaussian channel, the hard receiver detects the sign of each output and takes the majority, while the soft ML decoder adds the outputs. For the repetition code of length 3, the error probability of the hard receiver is

(a) larger than that of the soft decoder. ⋆
(b) smaller than that of the soft decoder.

> *Solution:* The answer is (a). By Section 2.3, the detector keeps the sign of each output and discards its magnitude, which the soft decoder uses, and ML on the full output has the smallest error probability.

6. On the real Gaussian channel with soft decisions, sending a symbol n times instead of once has the same error probability as sending it once with the SNR multiplied by

(a) n. ⋆
(b) $\sqrt{n}$.

Solution: The answer is (a). By Problem 2.5, the error probability is $\frac{1}{2}\operatorname{erfc}\left(\sqrt{n\,\mathrm{SNR}/2}\right)$, because the sum of the outputs has n times the signal and n times the noise variance, so its SNR is n times that of one output.

7. On the bit-erasing channel, the repetition code of length 3 is used and two of the three bits are erased. The ML decoder

(a) tosses a fair coin, because a majority of the bits is missing.
(b) decides the bit that arrived, and is right for sure. ⋆

Solution: The answer is (b). By Example 2.4(c), a bit that arrives is never wrong and only one codeword agrees with it, so the decoder is certain. It has to guess only when every bit is erased.

8. The repetition code is used on the bit-erasing channel and the two messages are not equiprobable. The MAP decoder differs from the ML decoder

(a) only when every bit is erased. ⋆
(b) whenever some bit is erased.

Solution: The answer is (a). By Proposition 2.3(c), when some bit arrives only one codeword has a nonzero likelihood and both decoders choose it; when every bit is erased the likelihoods are equal, ML tosses a coin and MAP chooses the message with the larger prior.

9. On the exponential noise channel, the noise is nonnegative and only adds. The repetition code uses the symbol 0 for one message and a positive symbol for the other. The ML decoder can make an error only when the message sent is the one with

(a) the symbol 0. ⋆
(b) the positive symbol.

Solution: The answer is (a). By Problem 2.6, the noise can raise every output of the zero codeword to the level of the positive symbol, and it can never lower an output of the positive codeword below it, so that codeword is always decoded correctly.

10. On the bit-flipping channel with flip probability p, where $\frac{1}{2} < p < 1$, the repetition code of length 3 is used and the receiver observes 011. The ML decoder decides the bit

(a) 0. ⋆
(b) 1.

Solution: The answer is (a). By Example 2.4(b), the likelihood ratio of 111 against 000 is $\big((1-p)/p\big)^{2d-3}$ with $d = 2$ ones, that is, $(1-p)/p$, which is below 1 when $p > \frac{1}{2}$: a flip is now more likely than no flip, so 000 is the more likely codeword and the ML decoder is the minority vote. Equivalently, flip every received bit and take the majority.